
A HOLISTIC EVALUATION OF 2D MOBILE POSE ESTIMATION MODELS FOR HOME-BASED REHABILITATION

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STATEMENT OF AUTHORSHIP

I hereby confirm that the work presented in this bachelor thesis has been performed and interpreted solely by myself except where explicitly identified to the contrary. I declare that I have used no other sources and aids other than those indicated. This work has not been submitted elsewhere in any other form for the fulfilment of any other degree or qualification.

Bonn, May 18, 2026

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ABSTRACT

This thesis evaluates three mobile pose estimation models—MediaPipe, MoveNet, and YOLOv8-Pose—plus six family variants for home-based rehabilitation. Standard benchmarks such as COCO and MPII cover general human pose estimation and do not reflect rehabilitation-specific conditions: fixed camera views, rehabilitation-style exercises, temporally stable predictions on video, and occasional secondary-person interference. We address this gap using REHAB24-6, a rehabilitation-oriented proxy benchmark of ten healthy adult volunteers performing six physiotherapist-guided exercises with synchronized RGB and motion-capture recordings.

The evaluation covers six dimensions—body-joint accuracy, viewpoint sensitivity, multi-person behavior, normalized frame-to-frame prediction displacement, detection completeness, and CPU inference speed—plus an auxiliary cross-dataset comparison on COCO val2017 under a different evaluation protocol. REHAB24-6 results are aggregated to ten subject-level clusters and analyzed with paired permutation tests, Holm correction, and bootstrap confidence intervals. MoveNet MultiPose has the lowest cluster-level body-joint error and remains significantly ahead of both alternatives after Holm correction. It also shows the lowest prediction-displacement value and the highest CPU throughput of the three main models. MediaPipe has the lowest catastrophic-failure rate on a coach extreme-case subset with a visible therapist, and YOLOv8 detects the full 12-joint skeleton most often. Detection counts alone do not explain coach-scene robustness; the three models form three architectural profiles, not a single scalar ranking. Each model is also dominated by a distinct failure mode consistent with its architectural paradigm. Model rankings on REHAB24-6 do not fully carry over to COCO under the protocol used here.

For deployment, MoveNet MultiPose is the strongest default for CPU-based body-joint rehabilitation analysis. Other configurations are better choices when detection completeness, lower coach-scene failure rate, richer native landmarks, or maximum throughput matter more. The most important next step toward clinical use is validation on real patient data.

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1 INTRODUCTION

1.1 MOTIVATION

Home-based physiotherapy depends on patients performing exercises correctly and regularly outside the clinic. Adherence is a persistent problem: prior work reports substantial non-adherence with physiotherapy treatment and exercise programs [Jac+10]. Non-adherence prolongs recovery, increases healthcare costs, and reduces clinical outcomes. Telerehabilitation aims to improve access and adherence through remote monitoring and feedback [SRR24].

Human pose estimation (HPE) offers a practical basis for such feedback. By estimating body-joint positions from RGB images, smartphone-based systems can monitor exercise execution without marker-based motion capture or other specialized equipment [Hel+21]. That reduces cost and equipment overhead for home-based rehabilitation.

Among the practical model families for this setting are MediaPipe Pose, MoveNet, and YOLOv8-Pose. They represent different architectural trade-offs in accuracy, speed, and robustness. Standard benchmarks do not cover these conditions, so the comparison has to be made on rehabilitation-style data.

1.2 PROBLEM STATEMENT

Standard pose estimation benchmarks such as COCO [Lin+14] and MPII [And+14] use diverse but generally favorable imagery. They are valuable for architecture comparison but do not cover the constraints of rehabilitation-oriented deployment:

- **Camera positioning.** Patients exercising at home often do not position themselves frontally to the camera, producing oblique or lateral body orientations that may increase self-occlusion.

- **Multi-person interference.** Family members, caregivers, or a supervising therapist may enter the camera’s field of view, requiring the model to correctly identify and retain the target subject.
- **Temporal stability.** Applications that provide real-time movement feedback depend on temporally stable predictions. Frame-to-frame instability degrades the user experience even when average accuracy is acceptable.
- **Resource constraints.** Patients typically lack dedicated GPU hardware, so models must achieve usable throughput on CPU.

Existing evaluations cover these constraints only partially. In the literature reviewed in Chapter 2, MoveNet has not been compared on a rehabilitation-oriented benchmark with motion-capture ground truth. Prior work also does not combine body-joint accuracy, viewpoint sensitivity, multi-person behavior, normalized frame-to-frame prediction displacement, intra-family speed–accuracy trade-offs, and cross-dataset ranking transfer in one mobile-model evaluation. This thesis addresses that gap by evaluating three main models and six variants on REHAB24-6 and comparing their rankings on COCO val2017 under a different evaluation protocol.

1.3 RESEARCH QUESTIONS

The thesis addresses five research questions:

- RQ1:** How accurate are mobile pose estimation models on a rehabilitation-oriented benchmark with motion-capture ground truth, and how complete are their keypoint detections on the shared body-joint skeleton?
- RQ2:** How temporally stable are their predictions, measured through normalized frame-to-frame prediction displacement?
- RQ3:** How robust are the models under suboptimal conditions, specifically body orientation relative to the camera and multi-person interference?
- RQ4:** How far do rankings obtained on the rehabilitation-oriented benchmark transfer to a standard large-scale benchmark, given that the two evaluations share the 12-joint skeleton but use dataset-specific protocols?

RQ5: What practical trade-offs and recommendations follow for mobile rehabilitation-oriented deployment?

1.4 CONTRIBUTIONS

The main contributions are:

1. **Rehabilitation-oriented evaluation of three mobile model families.** We evaluate MediaPipe, MoveNet, and YOLOv8 on REHAB24-6 under a shared protocol and extend the comparison to six additional family variants.
2. **Multi-metric benchmark.** We evaluate body-joint accuracy, normalized frame-to-frame prediction displacement, viewpoint sensitivity, multi-person behavior, detection completeness, and CPU inference speed within one comparison framework.
3. **Subject-cluster inference.** We aggregate REHAB24-6 results to subject-level clusters and report paired permutation tests with Holm correction and bootstrap confidence intervals; frame-level and video-level analyses are kept as sensitivity checks.
4. **Cross-dataset transferability analysis.** We compare all nine variants on COCO val2017 under an auxiliary protocol and quantify how rankings shift between the rehabilitation-oriented benchmark and a standard benchmark.
5. **Decomposition of multi-person behavior.** We analyze multi-person behavior through two complementary subsets: a segmentation-defined multi-person subset covering the full benchmark and a coach extreme-case subset of five c17 recordings with a prominently visible therapist.
6. **Failure-mode taxonomy.** We introduce a four-category taxonomy of per-frame failures—Missing-Detection, Confidence-Collapse, Multi-Person-Confusion, Keypoint-Displacement—and show that each main model is dominated by a distinct category consistent with its architectural paradigm.
7. **Deployment guidance.** We derive model-selection recommendations for CPU-based rehabilitation-oriented deployment from the combined accuracy, stability, robustness, completeness, and latency results.

1.5 THESIS STRUCTURE

The remainder of this thesis is organized as follows. Chapter 2 reviews human pose estimation, the evaluated model families, the evaluation metrics, and the related work. Chapter 3 defines the datasets, model configurations, metrics, filtering rules, and statistical procedures. Chapter 4 describes the evaluation pipeline. Chapter 5 reports the empirical findings. Chapter 6 interprets the results and derives practical implications. Chapter 7 summarizes the main findings and the next steps toward clinical validation.

2 BACKGROUND AND RELATED WORK

Three areas are relevant for this thesis: architectural paradigms, accuracy and temporal-stability metrics, and rehabilitation-oriented prior work. This chapter reviews each and positions the research gap.

2.1 HUMAN POSE ESTIMATION

Human pose estimation (HPE) localizes anatomical keypoints—joints and body landmarks—from visual input [Zhe+23]. For this thesis, two distinctions are useful: keypoint-based methods versus mesh-based methods [Gao+25], and 2D image-coordinate prediction versus 3D world-coordinate estimation. Mobile rehabilitation systems typically favor lightweight keypoint-based models, and this evaluation is strictly 2D, comparing predictions against projected motion-capture ground truth.

For rehabilitation, markerless HPE can support movement analysis without the cost and setup effort of marker-based systems [Hel+21; Deb+22]. The reliability of current mobile models under rehabilitation-oriented conditions remains insufficiently characterized.

2.2 ARCHITECTURAL PARADIGMS

The model comparison in this thesis is organized around three common architectural paradigms in modern HPE [Zhe+23].

TOP-DOWN APPROACHES first localize persons through bounding boxes or region proposals and then estimate pose within each detected region [TS14; NYD16; XWW18; Sun+19]. Their cost scales with the number of detected persons, and their accuracy depends on the reliability of the upstream detector.

BOTTOM-UP APPROACHES detect keypoints for the full image in one pass and then group them into individual persons. OpenPose [Cao+21] is the canonical representative. Bottom-up methods have a fixed computational cost with respect to person count, which is attractive for multi-person scenes. Their accuracy depends on the robustness of the grouping step, especially when keypoints from different persons are spatially close.

ONE-STAGE APPROACHES combine person detection and keypoint estimation in a single forward pass, typically by extending an object detector with a pose head. This often yields a favorable speed–accuracy trade-off for real-time use. A recent example is YOLOv8-Pose [JCQ23].

The three evaluated model families span all three paradigms: MediaPipe is top-down, MoveNet is bottom-up, and YOLOv8-Pose is one-stage.

2.3 EVALUATED MODELS AND VARIANTS

The evaluation focuses on three main models—one from each architectural paradigm—and adds six family variants to examine speed–accuracy trade-offs within each family. All configurations target mobile or edge deployment.

2.3.1 MEDIAPIPE POSE (BLAZEPOSE)

MediaPipe Pose [Baz+20] is a top-down pose estimation model developed within the MediaPipe framework [Lug+19]. It predicts 33 pose landmarks per detected person, including additional facial and distal-extremity landmarks beyond the 17 COCO keypoints used by MoveNet and YOLOv8-Pose. Three variants are available—Lite, Full, and Heavy—offering different accuracy–speed trade-offs under the same TFLite runtime.

2.3.2 MOVENET

MoveNet [Goo21] is a lightweight pose estimation model optimized for real-time inference on mobile and edge devices. It predicts the 17 COCO body keypoints and is available in both single-person and multi-person configurations. The MultiPose Lightning variant is used as the main MoveNet configuration because it exposes up to six persons per frame together with per-person bounding boxes; the family analysis additionally includes the SinglePose Lightning and SinglePose Thunder variants.

2.3.3 YOLOv8-Pose

YOLOv8-Pose [JCQ23] extends the YOLO object detection framework with a pose estimation head, producing bounding boxes and 17 COCO keypoints in a single forward pass. The Nano variant is used as the main configuration because it is the mobile-oriented entry point; the Small and Medium variants are additionally included for family-level speed–accuracy characterization. Unlike the TFLite-based MediaPipe and MoveNet pipelines, YOLOv8-Pose uses the Ultralytics PyTorch runtime.

2.4 EVALUATION METRICS AND BENCHMARKS

This thesis uses NMPJPE as its primary accuracy metric and normalized frame-to-frame prediction displacement as its primary stability proxy.

METRICS. The most common evaluation metrics are:

- **Percentage of Correct Keypoints (PCK):** the fraction of predicted keypoints within a threshold distance of the ground truth, typically normalized by head size or torso length [And+14].
- **Object Keypoint Similarity (OKS):** a per-keypoint similarity score with keypoint-specific tolerances, used as the primary metric in the COCO benchmark [Lin+14].
- **Mean Per Joint Position Error (MPJPE):** the mean Euclidean distance between predicted and ground-truth keypoint positions, usually reported in pixels or millimeters [Ion+14].
- **Normalized MPJPE (NMPJPE):** MPJPE normalized by a reference body segment length. This thesis uses NMPJPE as its primary accuracy metric (Section 3.5) because it provides a scale-normalized error on the shared 12-joint skeleton.

Video-based deployment also requires temporal stability [Zen+22]. This thesis therefore adds a frame-to-frame prediction-displacement metric, normalized by a reference body segment, to compare the movement of unsmoothed predicted trajectories.

STANDARD BENCHMARKS. COCO Keypoint Detection [Lin+14] and MPII Human Pose [And+14] dominate 2D HPE evaluation. For controlled 3D evaluation, Human3.6M [Ion+14] remains the canonical motion-capture benchmark.

LIMITATIONS FOR REHABILITATION-ORIENTED DEPLOYMENT. Standard benchmarks do not combine fixed-camera exercise recordings, repeated recordings per subject, multi-person interference requiring target-subject retention, and explicit temporal-stability analysis on video. This is why the thesis uses a rehabilitation-oriented proxy benchmark.

2.5 RELATED WORK

The related work below focuses on rehabilitation relevance, viewpoint sensitivity, and mobile-model comparison.

2.5.1 HPE IN REHABILITATION

Debnath et al. [Deb+22] survey computer-vision-based rehabilitation assessment and identify the scarcity of suitable public rehabilitation datasets as a central bottleneck. Černek et al. [ČSB25] introduce REHAB24-6 and evaluate selected pose-estimation methods on it, including MediaPipe and several YOLOv8-Pose variants, but not MoveNet, normalized frame-to-frame prediction displacement, coach-scene failure behavior, or the combined multi-person decomposition used here. Rode et al. [Rod+25] evaluate a broader set of models for clinical movement analysis, but emphasize per-joint accuracy rather than multi-person behavior, temporal prediction-displacement behavior, or cross-dataset transferability. Ullah et al. [Ull+25] present a MediaPipe-based physiotherapy scoring system as an application case rather than a comparative model study.

2.5.2 VIEWPOINT EFFECTS ON HPE

Aguilar-Ortega et al. [Agu+23] present the UCO Physical Rehabilitation dataset with multiple camera viewpoints and report a substantial increase in mean error from frontal to lateral views, but use server-grade rather than mobile-optimized models. Baldinger et al. [BRS25] investigate four camera viewing angles for OpenPose during lunge exercises and report higher errors under lateral viewpoints. Together, these studies establish that viewpoint affects HPE accuracy, but none evaluates mobile-optimized models on a rehabilitation-oriented benchmark with a matched viewpoint analysis.

2.5.3 MOBILE HPE COMPARISONS

Several studies compare mobile-optimized HPE models, usually on standard benchmarks rather than rehabilitation-oriented data. Jo and Kim [JK22] compare OpenPose, PoseNet, and MoveNet on mobile devices, and Chung et al. [COL22] provide a broader comparison with a similar emphasis on computational efficiency. Roggio et al. [Rog+24] review machine-learning pose-estimation models and consider MediaPipe and MoveNet suitable for remote fitness and rehabilitation applications, but do not quantify degradation under occlusion or crowded scenes. Hii et al. [Hii+23] validate MediaPipe for automated gait analysis without comparing it against alternative mobile models.

These studies provide useful reference points, but they do not combine rehabilitation-oriented data, temporal stability, and explicit target-subject selection behavior in multi-person scenes.

2.6 RESEARCH GAP

Table 1 summarizes how prior work covers the evaluation dimensions most relevant to mobile rehabilitation-oriented HPE.

TABLE 1.: Coverage of evaluation dimensions in related work. Checkmarks indicate that a study addresses the dimension; dashes indicate that it does not; “Part.” indicates partial coverage, for example a single model family, a restricted exercise set, or inference speed without accompanying deployment-oriented accuracy.

Study	Mobile	Rehab	Viewpoint	Multi-person	Stability	Variants	Cross-dataset
Aguilar-Ortega et al. [Agu+23]	–	✓	✓	–	–	–	–
Baldinger et al. [BRS25]	–	–	✓	–	–	–	–
Černek et al. [ČSB25]	Part.	✓	–	–	–	Part.	–
Rode et al. [Rod+25]	Part.	✓	–	–	–	–	–
Jo & Kim [JK22]	✓	–	–	–	–	–	–
Ullah et al. [Ull+25]	✓	Part.	–	–	–	–	–
This work	✓	✓	✓	✓	✓	✓	✓

The literature leaves six gaps that are relevant for this thesis.

- 1. MoveNet is missing from rehabilitation-oriented benchmarks.** Neither Černek et al. nor Rode et al. include MoveNet despite its relevance for mobile deployment.

2. **Temporal stability is rarely reported as a video metric.** Existing rehabilitation-oriented studies report per-frame accuracy and, at most, video-level averages, without measuring frame-to-frame prediction displacement.
3. **Multi-person behavior is treated as a single regime.** Multi-person scenes are recognized as difficult [Rog+24], but prior work does not decompose them into a segmentation-defined subset and a coach-scene extreme case, which capture different failure modes.
4. **Intra-family speed–accuracy trade-offs are only partially covered.** Prior work typically reports one variant per model family, so within-protocol comparisons across all three main families are not available.
5. **Cross-dataset transferability is often assumed rather than tested.** We found no study that directly compares ranking positions between a rehabilitation-oriented benchmark and a standard benchmark using a harmonized body-joint skeleton, even where the dataset-specific protocols differ.
6. **Repeated recordings per subject are rarely modeled at the inferential level.** Existing studies report frame-level or video-level summaries, so the dependency structure of repeated measurements is only partially addressed.

This thesis addresses these gaps by combining a rehabilitation-oriented evaluation on REHAB24-6 with an auxiliary transferability comparison on COCO val2017 and by reporting the primary inferential results at subject-cluster level.

3 METHODOLOGY

The study is motivated by home-based rehabilitation, but it is designed as a controlled model comparison on a rehabilitation-oriented proxy benchmark, not as direct clinical validation.

3.1 DATASET: REHAB24-6

The primary benchmark is REHAB24-6 [ČSB25], a public dataset of physiotherapist-guided rehabilitation exercises with synchronized RGB and motion-capture recordings. The released dataset contains 65 recordings of six exercises performed by 10 healthy adult volunteers, segmented into 1,072 annotated repetitions across two 30 fps RGB views. We use the projected 2D coordinates of the derived 26-joint motion-capture skeleton as ground truth for 2D pose evaluation.

REHAB24-6 is rehabilitation-oriented, but it is not a clinical patient cohort. We therefore treat it as a proxy testbed for rehabilitation-style movement analysis rather than as patient validation.

Table 2 summarizes the released dataset and the processed evaluation subset used here. In the repository state used for this thesis, the evaluation artifacts cover 63 recording identifiers with both camera views and matching ground truth, yielding 126 evaluated camera-view sequences.

Two aspects of the dataset are central to the later analyses. First, the fixed camera setup produces two characteristic viewpoints: camera c17 is predominantly frontal, whereas camera c18 is predominantly lateral. Second, the segmentation metadata labels whether additional persons are visible, enabling a dataset-wide multi-person subset analysis. In addition to this metadata-based subset, five c17 sequences with a prominently visible therapist are treated as an extreme-case coach subset.

TABLE 2.: *Overview of REHAB24-6 and the processed evaluation subset used in this thesis.*

Property	Value
Subjects	10 healthy adult volunteers (6 male, 4 female; age 25–50)
Exercises	6 physiotherapist-guided rehabilitation exercises
Public recordings	65
Annotated repetitions	1,072
RGB cameras	2 synchronized views at 30 fps
Ground truth	41-marker motion capture, projected to 2D; 26-joint skeleton
Processed evaluation subset	63 recording identifiers, 126 camera-view sequences
Frames analyzed	367,200 model-frame observations before filtering
Coach extreme-case subset	5 camera-c17 sequences

3.2 AUXILIARY BENCHMARK: COCO val2017

To assess how benchmark rankings transfer across domains, we additionally evaluate all model variants on COCO val2017 [Lin+14] for body-joint accuracy and ranking comparison only; rotation, temporal stability, and coach-scene analyses remain specific to REHAB24-6.

We evaluate 1,519 validation images that contain at least one annotated person with at least six labeled keypoints. For each image, the predicted instance is matched to the ground-truth person with the nearest torso center. The same 12 comparable body joints used in REHAB24-6 are evaluated, and only visible ground-truth joints are included. Because target matching and visibility handling differ from REHAB24-6, COCO serves as a transferability benchmark under a different evaluation protocol, not as a directly equivalent rehabilitation benchmark.

3.3 MODELS UNDER EVALUATION

The evaluation centers on three mobile pose-estimation models representing the three dominant architectural paradigms in human pose estimation [Zhe+23]: top-down, bottom-up, and one-stage. The primary comparison uses MediaPipe Full, MoveNet MultiPose, and YOLOv8 Nano. To analyze intra-family trade-offs, six additional variants are included: MediaPipe Lite and Heavy, MoveNet SinglePose Lightning and Thunder, and YOLOv8 Small and Medium.

TABLE 3.: *Model families, primary configurations, and additional variants.*

Family	Primary configuration	Additional variants	Native key-points	Native person output
MediaPipe Pose	Full	Lite, Heavy	33 landmarks	Up to 5 persons, no bounding boxes
MoveNet	MultiPose	SinglePose Lightning, SinglePose Thunder	17 COCO keypoints	MultiPose: up to 6 persons with bounding boxes; SinglePose: one person
YOLOv8-Pose	Nano	Small, Medium	17 COCO keypoints	Variable number of persons with bounding boxes

MEDIAPIPE POSE. The primary comparison uses the Full variant (`model_complexity=1`); the variant analysis additionally includes Lite and Heavy. For all MediaPipe variants, we use `min_pose_detection_confidence=0.1`, as stricter defaults caused avoidable detection failures in earlier runs. This parameter is a model-internal detection-stage threshold and should not be confused with the joint-level evaluation filter described in Section 3.9.

MOVE NET. MoveNet [Goo21] is evaluated in its MultiPose configuration for the main three-model comparison because it exposes multi-person detections and bounding boxes. The SinglePose Lightning and Thunder variants are included only for the family-level speed-accuracy analysis, since they do not expose comparable multi-person outputs.

YOLOv8-POSE. YOLOv8-Pose [JCQ23] uses the Nano model in the primary comparison, while the Small and Medium variants are included for intra-family trade-off analysis.

3.4 KEYPOINT MAPPING

The evaluated models and the motion-capture ground truth use different landmark definitions: MediaPipe predicts 33 landmarks, MoveNet and YOLOv8 use the 17-point COCO skeleton, and the ground truth provides 26 skeletal joints. We therefore map all predictions into 12 anatomically corresponding body joints: shoulders, elbows, wrists, hips, knees, and ankles.

TABLE 4.: Keypoint mapping across the ground truth, COCO skeleton, and MediaPipe landmarks.

Joint	Ground-truth joint	COCO index	MediaPipe index	Evaluated
Left Shoulder	LeftArm	5	11	✓
Right Shoulder	RightArm	6	12	✓
Left Elbow	LeftForeArm	7	13	✓
Right Elbow	RightForeArm	8	14	✓
Left Wrist	LeftHand	9	15	✓
Right Wrist	RightHand	10	16	✓
Left Hip	LeftUpLeg	11	23	✓
Right Hip	RightUpLeg	12	24	✓
Left Knee	LeftLeg	13	25	✓
Right Knee	RightLeg	14	26	✓
Left Ankle	LeftFoot	15	27	✓
Right Ankle	RightFoot	16	28	✓

Facial landmarks are excluded because the ground truth provides no corresponding markers. The resulting 12-joint evaluation space focuses the comparison on the major limb segments most relevant to rehabilitation-style movement analysis.

3.5 PRIMARY ACCURACY METRIC

The primary accuracy measure is the *Normalized Mean Per Joint Position Error* (NMPJPE), reported as a percentage of torso length [Ion+14]. For a frame with $N = 12$ evaluated joints,

$$\text{NMPJPE} = \frac{1}{N} \sum_{i=1}^N \frac{\|\mathbf{p}_i - \mathbf{g}_i\|_2}{L_{\text{torso}}} \times 100\% \quad (3.1)$$

where $\mathbf{p}_i$ and $\mathbf{g}_i$ are the predicted and ground-truth 2D coordinates of joint i . Torso length is defined from the shoulder and hip midpoints in the ground truth:

$$L_{\text{torso}} = \left\| \frac{\mathbf{g}_{\text{LSh}} + \mathbf{g}_{\text{RSh}}}{2} - \frac{\mathbf{g}_{\text{LHip}} + \mathbf{g}_{\text{RHip}}}{2} \right\|_2 \quad (3.2)$$

This normalization makes errors comparable across subjects, camera distances, and image resolutions. In addition to NMPJPE, we report *detection completeness* through the number of

valid evaluated joints per frame and the proportion of frames with all 12 joints present after confidence filtering.

3.6 ROTATION AND VIEWPOINT ANNOTATION

To characterize viewpoint sensitivity, we derive a camera-relative body orientation from the 3D ground truth rather than from the 2D predictions. This avoids circularity, since prediction errors would otherwise distort the angle estimate itself. The raw body orientation is computed from the 3D shoulder positions:

$$\theta_{\text{raw}} = \text{atan2}(|\Delta z|, |\Delta x|), \quad \Delta x = x_{\text{RSh}} - x_{\text{LSh}}, \quad \Delta z = z_{\text{RSh}} - z_{\text{LSh}} \quad (3.3)$$

Because the RGB cameras are not aligned with the motion-capture axes, a camera-relative offset must be applied. Since REHAB24-6 does not provide full camera extrinsics, we estimated this offset empirically from reference frames in which subjects stood approximately frontal to each camera:

$$\theta_{\text{c17}} = |\theta_{\text{raw}} - 65^\circ|, \quad \theta_{\text{c18}} = 90^\circ - \theta_{\text{c17}} \quad (3.4)$$

The 65° c17 offset is a single global calibration derived from three frontal reference recordings (PM_114, PM_122, PM_109), with an estimated uncertainty of approximately $\pm 5^\circ$. It is not re-estimated per video. The 10° angle bins used later in the rotation analysis are wider than this uncertainty, so the calibration error stays inside the binning resolution.

The resulting values range from 0° (frontal) to 90° (lateral) and are analyzed in 10° bins. In practice, the REHAB24-6 distribution is strongly bimodal: frames are concentrated in mostly frontal or mostly lateral views, with comparatively few intermediate poses. We therefore interpret this analysis as viewpoint sensitivity under fixed camera geometry, not as a dense continuous rotation study.

3.7 MULTI-PERSON CATEGORIZATION AND PERSON SELECTION

Multi-person effects are examined through two complementary subsets. First, the segmentation metadata provides an extra_person severity label, which allows a dataset-wide multi-person

subset analysis. We define this subset as frames with severity ≥ 2 , corresponding to clearly visible secondary persons or body parts. Second, we define an extreme-case coach subset of five c17 sequences (PM_010, PM_011, PM_108, PM_119, PM_121), identified by manual review for sustained therapist visibility across the recording.

When more than one person is detected, the evaluation pipeline must select a target instance. We therefore use model-specific heuristics that match the native outputs of each model.

For MoveNet MultiPose and YOLOv8, the selected instance is the one with the largest bounding-box area. For MediaPipe, which does not provide native bounding boxes, the selected instance is the one with the largest torso extent in image space. This avoids pseudo-bounding boxes derived from keypoint extremes, which vary strongly with limb spread and do not reliably reflect apparent person size.

These heuristics are evaluation design choices matching each model’s native output format. We use them to keep the comparison reproducible and output-consistent, and we interpret selection effects separately from the underlying pose predictions: selection belongs to the evaluation pipeline, not to the pose estimator itself. Later analyses distinguish between detection-level behavior, which determines how often multiple candidate persons reach the selection stage, and downstream behavior beyond simple person-count frequency.

3.8 TEMPORAL STABILITY

Temporal stability is approximated through normalized frame-to-frame prediction displacement. For two consecutive sampled frames t and $t + 1$, the displacement of joint i is

$$D_i^t = \frac{\|\mathbf{p}_i^{t+1} - \mathbf{p}_i^t\|_2}{\frac{1}{2}(L_{\text{torso}}^t + L_{\text{torso}}^{t+1})} \times 100\% \quad (3.5)$$

where the denominator is the mean torso length of the two corresponding ground-truth frames. A joint contributes only if it is valid in both frames after confidence filtering. Per-frame displacement is the mean across valid evaluated joints, and per-model displacement is the mean across all such frame pairs. This metric does not subtract ground-truth motion; it is therefore a stability proxy for the predicted trajectories rather than a direct decomposition of model noise from true movement. Because the source frame pairs come from the same videos

under the shared sampling protocol, lower values indicate less frame-to-frame movement in the predicted trajectories.

3.9 CONFIDENCE FILTERING AND OUTLIER HANDLING

All evaluated models provide a confidence value per predicted joint. For accuracy computation, joints with confidence below 0.5 are excluded from the frame-level NMPJPE. This threshold is deliberately applied at the joint level rather than as a whole-frame rejection rule, so partially reliable predictions remain usable. The same threshold is used for temporal stability, where a joint contributes only if it is valid in both consecutive frames.

Frames in which no valid person is detected are marked as detection failures and excluded from central-tendency accuracy summaries. Frames with NMPJPE exceeding 100% of torso length are treated as extreme outliers and interpreted as catastrophic misassignment or wrong-person failures. These outliers are removed from the primary central-tendency subset, but they remain relevant for robustness analyses in which failure rate itself is an outcome.

Accordingly, the thesis distinguishes between two reporting regimes:

- **Central tendency:** valid subset only (no detection failure, no >100% outlier, no MoCap error on REHAB24-6).
- **Failure behavior:** outliers retained when the analysis explicitly concerns catastrophic person-switch events or coach-scene robustness.

3.10 FRAME-INDEPENDENT EVALUATION MODE

All models are evaluated frame-independently. MediaPipe runs in IMAGE mode, MoveNet processes each frame without temporal cropping or tracking memory, and YOLOv8 is evaluated through frame-wise prediction rather than object tracking. This choice isolates model-level prediction behavior from pipeline-level temporal engineering such as tracker association, Kalman filtering, or smoothing.

This design has an important consequence for interpretation: the reported person-switch and temporal-stability values describe unsmoothed model behavior under frame-independent processing. They should therefore be read as upper-bound failure characteristics of the raw model outputs, not as direct estimates of a fully engineered deployment pipeline.

3.11 INFERENCE TIME BENCHMARK

We benchmark inference time separately from the accuracy evaluation to isolate model latency from data loading and metric computation. All benchmarks run on CPU-only hardware and use three c17 videos selected consistently across models, with 500 measured frames per video and a 50-frame warm-up period. Timing is recorded per frame using high-resolution performance counters.

The benchmark covers all nine evaluated configurations. Each model is measured sequentially to avoid resource contention. For deployment realism, MediaPipe variants use the Tasks API, MoveNet variants use TFLite runtimes, and YOLOv8 variants use the Ultralytics PyTorch runtime.

3.12 FRAME SAMPLING AND STATISTICAL ANALYSIS

The RGB videos are recorded at 30 fps. To reduce temporal redundancy and the amount of near-duplicate frames, we process every third frame, yielding an effective sampling rate of 10 Hz. This is sufficient for the relatively slow rehabilitation-style movements in the dataset while substantially reducing autocorrelation and computational cost.

For inferential analysis on REHAB24-6, frames are not treated as independent observations. Instead, we first compute a median NMPJPE per camera-view sequence, then average these sequence medians within each subject cluster identified by the segmentation metadata `person_id`. This yields 10 primary subject clusters in the processed repository state. Pairwise model comparisons on this primary level use two-sided sign-flip permutation tests on the cluster-wise differences, evaluated by exact enumeration over all $2^{10} = 1024$ sign assignments. We accompany the tests with percentile bootstrap confidence intervals (20,000 resamples) and Holm correction for multiple comparisons. Video-level comparisons are reported only as a secondary sensitivity analysis and use the same paired test with Monte Carlo sampling because the video-level sign-assignment space is much larger.

This subject-cluster aggregation is used only for REHAB24-6 inferential statements. Cross-dataset comparisons with COCO remain descriptive because the COCO protocol differs in target matching and visibility handling.

4 IMPLEMENTATION

The implementation separates expensive model inference from downstream metric computation so that filtering, aggregation, or statistical choices can be revised without repeating inference.

4.1 PIPELINE OVERVIEW

The evaluation uses a two-phase architecture. Phase 1 runs inference once per model on all REHAB24-6 recordings and on the COCO val2017 subset, storing per-frame keypoint predictions and confidence values as compressed `.npz` arrays. Phase 2 loads these predictions together with the matching ground truth, applies the filtering rules defined in Chapter 3, and computes all reported metrics and statistical tests. Figures and tables read exclusively from the Phase 2 CSV and JSON artifacts, which keeps each reported number traceable to a single canonical source.

All nine evaluated configurations—the three main models and the six family variants—share the same Phase 1 and Phase 2 code. Each variant plugs into the same estimator interface; model weights, keypoint counts, and person outputs differ, but the downstream code does not branch on model type.

4.2 MODEL ESTIMATOR INTERFACE

The three model families expose fundamentally different APIs: MediaPipe uses its Tasks API, MoveNet runs through TensorFlow Hub with manual padding and unpadding of the input tensor, and YOLOv8 uses the Ultralytics runtime. A common abstract `PoseEstimator` base class wraps these APIs so that Phase 1 can iterate over frames and models uniformly. Each concrete estimator loads its own weights, returns native candidate detections for the current frame, and converts native keypoint formats to the shared COCO index space. The evaluation pipeline

then applies the corresponding person-selection logic (Section 3.7) consistently across models before downstream metric computation. The abstraction is deliberately thin: it hides format differences but leaves evaluation logic, including selection, in Phase 2.

4.3 DATA ARTIFACTS AND REPRODUCIBILITY

Phase 1 stores predictions per model and per recording in a normalized `.npz` format used by Phase 2. The stored array has shape $(N, 17, 3)$ in the shared COCO index space, where N is the number of sampled frames for the recording, together with a length- N array of camera-relative rotation angles. Native outputs with different keypoint counts or coordinate conventions (in particular MediaPipe’s 33 landmarks) are mapped into this shared format by the estimator before writing. Phase 2 produces the result artifacts used throughout Chapters 5 and 6: per-frame evaluation tables, subject-level cluster statistics with permutation-based p -values and bootstrap confidence intervals, frame-to-frame prediction-displacement tables, coach-scene detection counts, hip-sensitivity tables, and the CPU inference benchmark.

Every headline number in the thesis is traceable to one of these artifacts; a separate provenance file maps each reported value to its source file. The inference benchmark runs in a dedicated sequential script with high-resolution timers and a fixed per-model sample budget, isolated from the main pipeline to avoid resource contention. Model weights are downloaded on first use, and Phase 2 is deterministic on the stored `.npz` files: given the same predictions and the same filtering and statistical settings, re-running Phase 2 reproduces the reported metrics exactly. Determinism across different Phase 1 runtimes is not separately guaranteed.

5 RESULTS

Unless stated otherwise, REHAB24-6 accuracy values refer to the valid subset defined in Chapter 3: no detection failure, no extreme outlier above 100% NMPJPE, and no MoCap annotation error. Inferential statements on REHAB24-6 use the subject-cluster aggregation described in Section 3.12; cross-dataset comparisons with COCO are descriptive.

5.1 DESCRIPTIVE STATISTICS

The primary three-model REHAB24-6 benchmark contains 126 processed camera-view sequences, corresponding to 367,200 frame-level observations before filtering, 363,074 after removing detection failures, and 354,879 in the valid central-tendency subset. Because confidence filtering and the >100% outlier cut are applied per model, the valid frame counts in Table 5 differ slightly between models (115,711 to 119,729). Five c17 sequences form the coach extreme-case subset. The viewpoint distribution remains strongly bimodal after filtering: on the valid MoveNet subset, 40.8% of frames fall into the frontal range (0–20°), 17.5% into the intermediate range (20–60°), and 41.6% into the lateral range (60–90°).

5.2 OVERALL ACCURACY

Table 5 reports the primary REHAB24-6 accuracy results on the valid subset. MoveNet achieved the lowest frame-level mean NMPJPE at 10.52%, followed by MediaPipe at 12.53% and YOLOv8 at 12.77%; the same ordering remained after subject-cluster aggregation.

TABLE 5.: Overall REHAB24-6 accuracy on the valid subset. Lower is better.

Model	Mean (%)	Median (%)	Std (%)	<i>n</i>
MoveNet	10.52%	9.85%	4.66%	119,439
MediaPipe	12.53%	11.26%	7.05%	119,729
YOLOv8	12.77%	11.23%	6.73%	115,711

The inferential comparison uses 10 subject clusters derived from the segmentation metadata. Table 6 shows the paired sign-flip permutation results with bootstrap confidence intervals and Holm correction. MoveNet outperformed both MediaPipe and YOLOv8 at the primary inferential level, and MediaPipe outperformed YOLOv8.

TABLE 6.: Pairwise REHAB24-6 accuracy comparisons on the primary subject-cluster level ($n = 10$). Differences are reported as model A minus model B in mean cluster-level NMPJPE.

Comparison	Mean difference	95% CI	p	p_{Holm}
MediaPipe – MoveNet	1.01%	[0.62, 1.42]	0.0020	0.0059
MediaPipe – YOLOv8	−0.49%	[−0.85, −0.12]	0.0352	0.0352
MoveNet – YOLOv8	−1.50%	[−1.71, −1.27]	0.0020	0.0059

Figure 1 shows the per-cluster distribution that underlies this test. The separation between MoveNet and the two alternatives is visible across most clusters, whereas MediaPipe and YOLOv8 show more overlap.

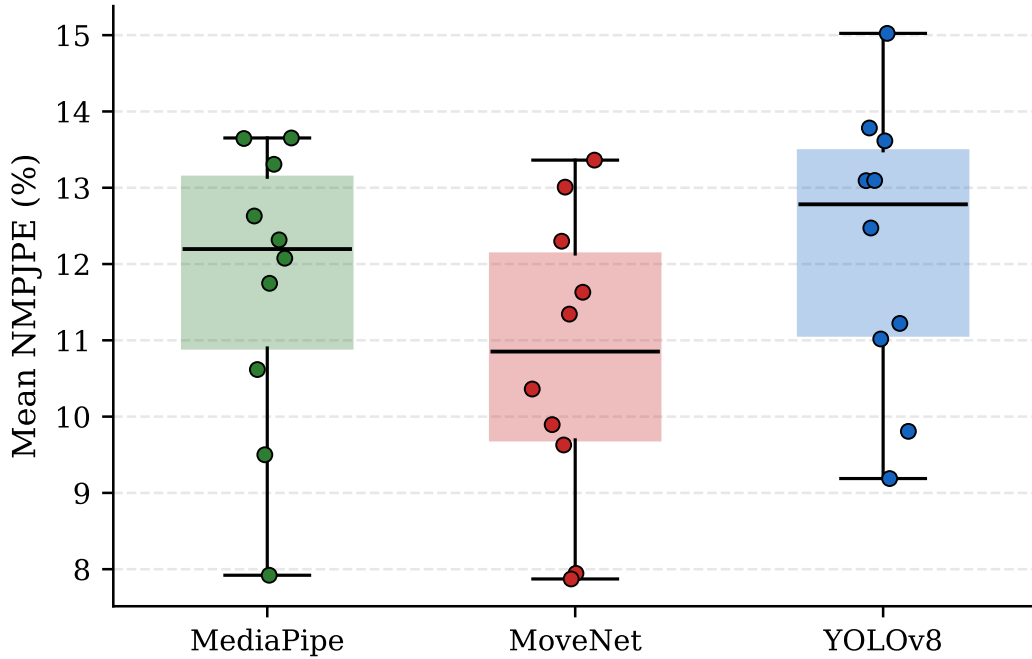


FIGURE 1.: Subject-cluster accuracy distributions on REHAB24-6. Each point represents one subject cluster defined by the segmentation metadata; boxplots summarize the distribution of the cluster-level mean of per-video median NMPJPE across the 10 clusters. Lower is better.

Per-exercise aggregation confirms that MoveNet ranks first on all six REHAB24-6 exercises (Appendix A.1).

5.3 MODEL VARIANTS

Table 7 extends the REHAB24-6 comparison to all nine model variants. MoveNet MultiPose achieved the lowest valid-subset NMPJPE. Within families, larger YOLOv8 variants improved over Nano, MediaPipe Heavy improved over Full and Lite, MoveNet SP_Lightning traded accuracy for a substantial speed gain, and MoveNet SP_Thunder traded accuracy without a meaningful speed advantage over MultiPose.

TABLE 7.: REHAB24-6 valid-subset accuracy for all nine evaluated variants. Lower is better.

Variant	Mean NMPJPE	Median NMPJPE	Valid frames
MoveNet_MultiPose	10.52%	9.85%	119,439
YOLOv8_Medium	10.86%	10.01%	114,591
YOLOv8_Small	11.43%	10.27%	116,009
MediaPipe_Heavy	11.51%	10.59%	119,630
MediaPipe_Full	12.53%	11.26%	119,729
YOLOv8_Nano	12.77%	11.23%	115,711
MoveNet_SP_Thunder	13.33%	11.93%	117,719
MediaPipe_Lite	13.54%	11.77%	119,392
MoveNet_SP_Lightning	14.97%	12.83%	94,395

5.4 CROSS-DATASET COMPARISON

Table 8 compares the variant rankings between REHAB24-6 and COCO val2017. On REHAB24-6, MoveNet MultiPose ranked first, whereas YOLOv8 Medium ranked first on COCO. When COCO is restricted to single-person images, the leading order of the strongest COCO variants is preserved and the ranking shift relative to REHAB24-6 persists.

Figure 2 visualizes these rank movements directly. The largest shifts affect the MoveNet family and the larger MediaPipe variants, while YOLOv8 changes least overall.

Among the three main models, MoveNet ranked first on both datasets, while YOLOv8 and MediaPipe swapped order.

TABLE 8.: Cross-dataset ranking comparison for all nine variants. “COCO single” refers to the valid subset restricted to images containing one annotated person.

Variant	REHAB rk.	REHAB %	COCO rk.	COCO %	COCO single %
MoveNet_MultiPose	1	10.52%	4	12.44%	12.55%
YOLOv8_Medium	2	10.86%	1	10.15%	10.53%
YOLOv8_Small	3	11.43%	2	10.88%	11.74%
MediaPipe_Heavy	4	11.51%	7	15.08%	14.50%
MediaPipe_Full	5	12.53%	8	17.21%	16.73%
YOLOv8_Nano	6	12.77%	6	13.69%	14.87%
MoveNet_SP_Thunder	7	13.33%	3	12.01%	11.84%
MediaPipe_Lite	8	13.54%	9	20.07%	19.33%
MoveNet_SP_Lightning	9	14.97%	5	12.63%	12.61%

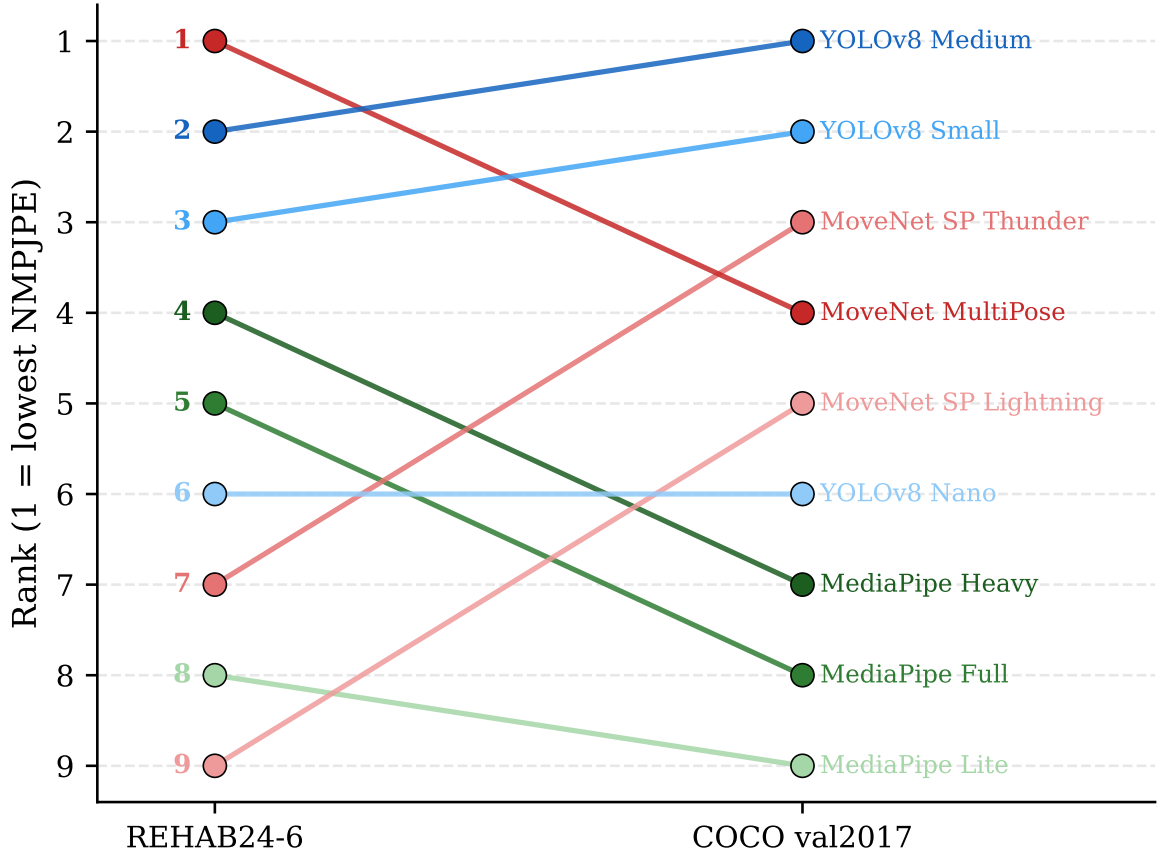


FIGURE 2.: Ranking shifts between REHAB24-6 and COCO val2017 across the nine evaluated variants. Rankings are based on mean NMPJPE within each dataset’s valid subset, with rank 1 indicating the lowest NMPJPE. Exact mean values are reported in Table 8.

5.5 VIEWPOINT SENSITIVITY

Table 9 summarizes the frontal-to-lateral viewpoint degradation on REHAB24-6. MoveNet exhibited the smallest increase from frontal to lateral views (+17.8%), followed by MediaPipe (+27.5%) and YOLOv8 (+31.7%).

TABLE 9.: REHAB24-6 viewpoint sensitivity on the valid subset. Frontal corresponds to 0–20° and lateral to 60–90°.

Model	Frontal mean	Lateral mean	Degradation
MoveNet	9.75%	11.49%	+17.8%
MediaPipe	11.28%	14.38%	+27.5%
YOLOv8	11.14%	14.66%	+31.7%

Figure 3 shows the full binned trend across 10° rotation bins. The uneven pattern in the intermediate bins reflects the fixed-camera geometry and strongly bimodal viewpoint distribution; the figure should not be read as a densely sampled continuous rotation study.

5.6 MULTI-PERSON BEHAVIOR

We analyze multi-person behavior in two subsets: a segmentation-defined subset covering the full benchmark and an extreme-case coach subset of five c17 views with a prominently visible therapist.

SEGMENTATION-DEFINED SUBSET. On the segmentation-based multi-person subset, the filtered mean error changed only slightly relative to the remaining valid frames. MediaPipe increased by 1.2%, while MoveNet and YOLOv8 were slightly lower on the multi-person subset than on the complementary subset.

TABLE 10.: REHAB24-6 segmentation-defined multi-person subset ($\text{severity} \geq 2$) on the valid data.

Model	Single-person mean	Multi-person mean	Change	Multi-person frames
MediaPipe	12.51%	12.67%	+1.2%	10,596
MoveNet	10.56%	10.04%	−5.0%	10,703
YOLOv8	12.82%	12.25%	−4.4%	10,286

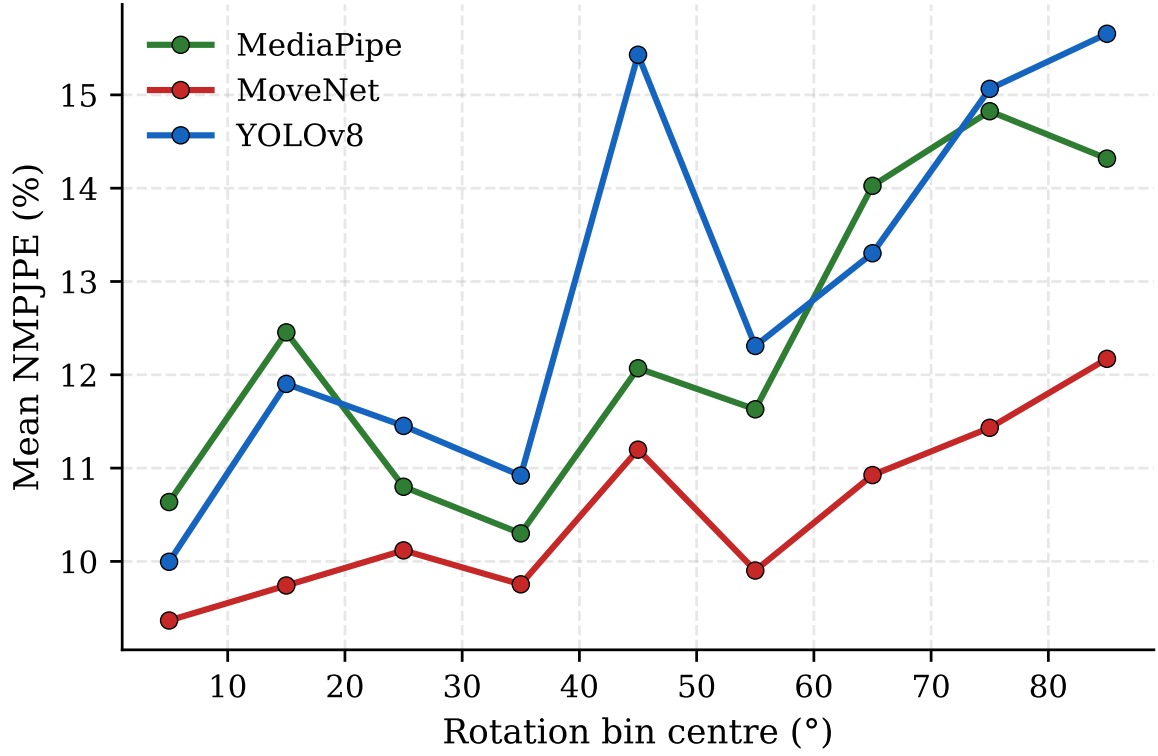


FIGURE 3.: Mean NMPJPE by 10° rotation bin on the REHAB24-6 valid subset. Points show binned means; the connecting lines visualize the viewpoint trend across bins. Under REHAB24-6’s fixed camera geometry and bimodal viewpoint distribution, this figure should be interpreted as a viewpoint comparison rather than as a dense continuous rotation study. Lower is better.

COACH EXTREME-CASE SUBSET. The five coach sequences show a different failure regime. Table 11 combines the raw detection-count analysis with the catastrophic outlier rate on the coach subset. MediaPipe detected two or more persons in 23% of sampled coach frames, MoveNet in 16%, and YOLOv8 in 62%; the corresponding coach-scene outlier rates were 9.1%, 13.8%, and 13.9%.

TABLE 11.: *Coach extreme-case subset: sampled secondary-person frequency, coach-scene outlier rate, and mean NMPJPE. Detection counts are based on every tenth frame across 1,178 sampled coach frames; outlier rates and mean NMPJPE are computed over all non-failure coach frames, with NMPJPE reported before the >100% outlier cut so that catastrophic misassignment events remain reflected in the mean.*

Model	2+ persons detected	Coach outlier rate	Mean coach NMPJPE
MediaPipe	23%	9.1%	45.4%
MoveNet	16%	13.8%	62.2%
YOLOv8	62%	13.9%	66.0%

The five coach videos differ substantially. Table 12 shows that PM_010-c17 is the only coach sequence in which all three models exhibit sequence-level mean NMPJPE above 65%, whereas MediaPipe stays below 40% in the other four coach sequences.

TABLE 12.: *Per-video mean NMPJPE on the five coach extreme-case sequences.*

Coach sequence	MediaPipe	MoveNet	YOLOv8
PM_010-c17	81.9%	70.7%	69.9%
PM_011-c17	38.0%	59.7%	66.2%
PM_108-c17	17.2%	46.2%	51.1%
PM_119-c17	24.1%	74.0%	73.0%
PM_121-c17	30.9%	60.3%	75.3%

Figure 4 illustrates the distinct model outputs on a coach extreme-case frame.

5.7 TEMPORAL STABILITY

Temporal stability is assessed through normalized frame-to-frame prediction displacement on valid frame pairs. MoveNet produced the lowest mean displacement at 3.81%, followed by YOLOv8 at 5.22% and MediaPipe at 6.03%.



FIGURE 4.: Illustrative outputs on a coach extreme-case frame (PM_119-c17, frame 105). MediaPipe (green) localizes the patient; MoveNet (red) detects patient and therapist but selects the therapist under the largest-area rule; YOLOv8 (blue) detects patient, therapist, and a partial border person but selects the patient. Keypoints are shown for all detections; native bounding boxes are drawn only for the selected candidate. Aggregate coach-scene frequencies are reported in Table 11.

Across all nine variants, the lowest mean displacement was observed for MoveNet_SP_Thunder (3.31%), followed by MoveNet_SP_Lightning (3.61%) and MoveNet_MultiPose (3.81%).

5.8 DETECTION COMPLETENESS

Table 13 summarizes detection completeness. YOLOv8 produced full 12-joint detections in 79.1% of frames, compared to 55.1% for MediaPipe and 38.2% for MoveNet.

TABLE 13.: Detection completeness on REHAB24-6 after removing full-frame detection failures.

Model	Full detection (12/12)	Mean valid joints	Median valid joints
MediaPipe	55.1%	10.72	12
MoveNet	38.2%	10.31	11
YOLOv8	79.1%	11.53	12

5.9 INFERENCE TIME AND SPEED–ACCURACY TRADE-OFFS

Table 14 reports the CPU-only inference benchmark for all nine evaluated variants. MoveNet_SP_Lightning was the fastest configuration at 100.6 FPS, whereas YOLOv8_Medium was the slowest at 4.6 FPS.

TABLE 14.: CPU-only inference benchmark for all nine evaluated variants.

Variant	Mean (ms)	Median (ms)	FPS	P95 (ms)
MoveNet_SP_Lightning	9.9	9.8	100.6	10.9
MoveNet_MultiPose	36.1	34.8	27.7	45.8
MoveNet_SP_Thunder	36.2	36.0	27.6	38.1
MediaPipe_Lite	45.7	42.2	21.9	70.9
YOLOv8_Nano	52.9	51.1	18.9	60.3
MediaPipe_Full	67.9	64.0	14.7	116.1
YOLOv8_Small	97.1	93.9	10.3	107.5
MediaPipe_Heavy	165.3	135.3	6.0	304.1
YOLOv8_Medium	215.4	211.1	4.6	246.3

Figure 5 combines REHAB24-6 valid-subset accuracy with the dedicated sequential CPU benchmark. The dashed frontier marks the non-dominated configurations under these two criteria.

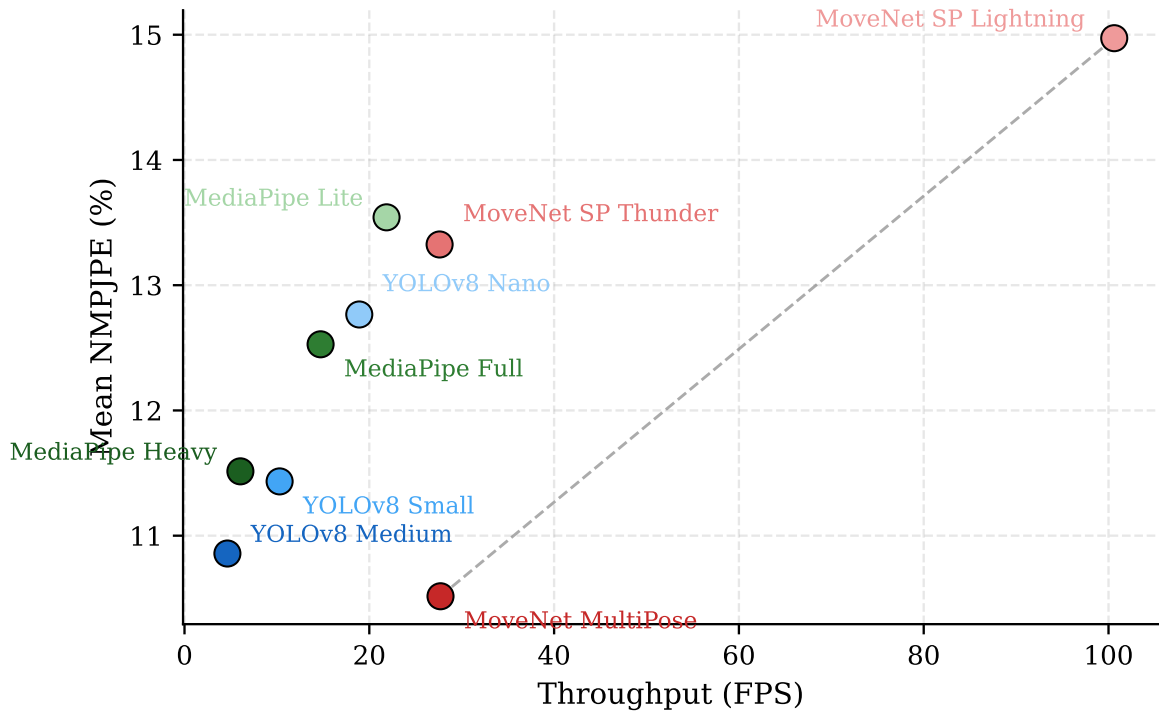


FIGURE 5.: Speed–accuracy trade-off across all nine evaluated variants. The horizontal axis shows throughput from the dedicated sequential CPU-only inference benchmark; the vertical axis shows mean NMPJPE on the REHAB24-6 valid subset. The dashed line marks the non-dominated configurations under these two criteria. Lower and further right is better.

Figure 5 together with Tables 7 and 14 shows that several family-level accuracy gains come at substantial latency cost, most clearly for MediaPipe Heavy and YOLOv8 Medium.

The reported throughput reflects the implemented native deployment stacks as well as the model variants themselves, not a backend-normalized architecture benchmark.

5.10 PER-JOINT ERROR PATTERNS

Table 15 summarizes the regional error profile across the 12 evaluated joints. MediaPipe shows the highest hip error among the three main models, while MoveNet yields the lowest error in the lower-body regions overall.

TABLE 15.: *Regional per-joint NMPJPE on the REHAB24-6 valid subset.*

Body region	MediaPipe	MoveNet	YOLOv8
Shoulders	8.76%	10.29%	10.21%
Elbows	11.24%	10.00%	13.08%
Wrists	12.01%	11.84%	16.25%
Hips	16.72%	11.28%	14.59%
Knees	9.92%	7.76%	9.87%
Ankles	14.59%	11.41%	12.59%

On the three main models, MediaPipe’s mean NMPJPE decreases from 12.53% to 11.66% when the hip joints are excluded, whereas MoveNet changes only marginally from 10.52% to 10.46%. The MediaPipe–MoveNet gap shrinks from 2.01 percentage points on all 12 joints to 1.21 percentage points on the 10 non-hip joints.

5.11 FAILURE-MODE DECOMPOSITION

The previous sections report separate failure indicators—detection failures, high-NMPJPE outliers, multi-person context, low valid-joint counts. We combine them into one mutually exclusive taxonomy that classifies every failure frame into one of four categories. A frame counts as a failure if it is flagged as a detection failure or as an NMPJPE outlier (>100% of torso length). The four categories are applied in order:

1. **Missing-Detection:** the model returned no detection for the frame.

2. **Confidence-Collapse:** detection succeeded, but fewer than six of the 12 evaluated joints remained after the joint-level confidence filter.
3. **Multi-Person-Confusion:** outlier with sufficient valid joints in a frame with a visible secondary person ($\text{extra_person_severity} \geq 2$).
4. **Keypoint-Displacement:** outlier with sufficient valid joints in a frame without a clearly visible secondary person—a catastrophic misprediction without an obvious detection- or selection-level cause.

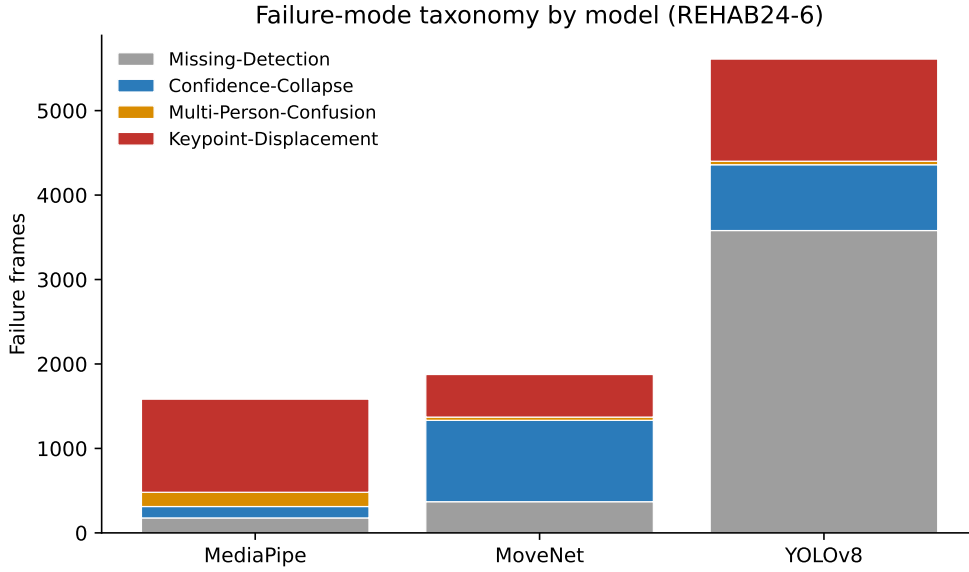


FIGURE 6.: Failure-mode distribution on REHAB24-6. Each bar represents one main model; the four segments give the absolute number of failure frames per category. Exact counts are reported in Table 17.

Each model is dominated by a different category. MediaPipe failures are predominantly Keypoint-Displacement (69.6% of 1,583 failures), MoveNet failures are predominantly Confidence-Collapse (51.4% of 1,875), and YOLOv8 has the largest absolute failure count, driven by Missing-Detection (63.8% of 5,610). YOLOv8 returned no prediction for 3,580 frames, roughly twenty times the MediaPipe rate. Multi-Person-Confusion accounts for at most 11% of failures across models.

The split is robust to the Confidence-Collapse threshold. Varying the valid-joint cutoff between 4 and 8 preserves the dominant category for each model.

6 DISCUSSION

The discussion separates direct empirical findings from narrower interpretations and clearly marked open questions.

6.1 ANSWERING THE RESEARCH QUESTIONS

RQ1: HOW ACCURATE AND COMPLETE ARE MOBILE POSE ESTIMATION MODELS IN THE BENCHMARK SETTING USED HERE? MoveNet achieved the lowest body-joint error among the three main models on REHAB24-6. This ranking holds both on the frame-level valid subset and on the primary subject-cluster aggregation. The paired cluster-level analysis further indicates that MoveNet outperformed both MediaPipe and YOLOv8 after Holm correction, while MediaPipe also outperformed YOLOv8. Detection completeness follows a different ordering: YOLOv8 produced full 12-joint detections most frequently, followed by MediaPipe and then MoveNet. Within the shared 12-joint benchmark space used here, the main comparison therefore separates accuracy from completeness rather than identifying one model as strongest on both dimensions.

RQ2: HOW STABLE ARE PREDICTIONS OVER TIME? MoveNet also produced the lowest normalized frame-to-frame prediction displacement, followed by YOLOv8 and MediaPipe. The same tendency extends to the broader variant set, where all three MoveNet variants occupy the lowest-displacement range. The stability result therefore supports the same overall pattern rather than introducing a separate trade-off.

RQ3: HOW ROBUST ARE THE MODELS UNDER SUBOPTIMAL CONDITIONS? The answer depends on the condition. For viewpoint change, MoveNet is the most robust of the three main models. For segmentation-defined multi-person frames, none of the models shows a large degradation after filtering. For the coach extreme-case subset, MediaPipe shows the lowest catastrophic-failure

rate. These are distinct robustness regimes and should not be collapsed into a single scalar notion of “multi-person robustness”.

RQ4: HOW FAR DO RANKINGS TRANSFER ACROSS BENCHMARKS? Transferability is limited rather than absent. Across the full nine-model set, the leading configuration changes between REHAB24-6 and COCO, and this shift persists when COCO is restricted to single-person images. Among the three main models, MoveNet remains strongest on both datasets, while MediaPipe and YOLOv8 swap order. Because the COCO comparison uses a different protocol from REHAB24-6, this result supports only bounded transferability under the present setup rather than a protocol-independent ranking claim.

RQ5: WHAT PRACTICAL TRADE-OFFS AND RECOMMENDATIONS FOLLOW FOR MOBILE REHABILITATION APPLICATIONS? The practical answer is use-case dependent, but the overall pattern is clear. For body-joint-only analysis under CPU constraints, MoveNet MultiPose is the strongest default choice because it combines the best REHAB24-6 accuracy with the best main-model temporal stability and substantially higher throughput than MediaPipe Full. YOLOv8 remains attractive when full 12-joint coverage is especially important. MediaPipe remains relevant when richer native landmarks, simpler integration, or lower catastrophic failure in the coach extreme-case subset matter more than the best body-joint accuracy.

6.2 THREE ARCHITECTURAL PROFILES

The three main models form three distinct architectural profiles rather than a simple best-to-worst ranking.

YOLOv8: HIGH-SENSITIVITY ONE-STAGE DETECTION. YOLOv8 returns complete 12-joint detections most frequently and exposes multiple persons in 62% of sampled coach frames. This makes it the most detection-complete of the three main models, but also the model most frequently forced into downstream person disambiguation under the coach extreme-case protocol. At the same time, it is less accurate than MoveNet on REHAB24-6 and less robust to viewpoint change.

MOVE.NET: LOW-EXPOSURE BOTTOM-UP PROFILE. MoveNet combines the strongest REHAB24-6 body-joint accuracy with the lowest main-model prediction displacement and substantially higher throughput than MediaPipe Full. In the coach subset it exposes multiple persons in only 16% of sampled frames, making it the most conservative model in terms of secondary-person frequency. Yet this conservative exposure does not translate into the lowest coach outlier rate, which is why MoveNet’s coach-scene behavior cannot be reduced to simple person-count frequency.

MEDIAPIPE: FILTERED TOP-DOWN PROFILE. MediaPipe occupies an intermediate position in raw secondary-person exposure but the lowest catastrophic-failure rate in coach scenes. It is slower than MoveNet and less accurate on the evaluated 12-joint body benchmark, but its coach-scene failure profile differs from both alternatives. It is therefore not the general winner, but it remains a distinct deployment profile with different strengths and weaknesses.

FAILURE-MODE SIGNATURE. Section 5.11 decomposes each model’s failure frames into four categories. The dominant category for each model is consistent with the design logic of its architectural paradigm: MediaPipe’s top-down commit-then-predict pipeline aligns with Keypoint-Displacement (confident misprediction in otherwise clean frames); MoveNet’s bottom-up per-keypoint confidence aligns with Confidence-Collapse (graceful degradation with many joints filtered out); YOLOv8’s one-stage coupling of detection and pose aligns with Missing-Detection (no subject, no pose). This correspondence is observational rather than mechanistic. Each paradigm is represented by a single model family in our evaluation, so we cannot separate paradigm-level effects from training data, confidence-calibration choices, or landmark definitions. We note the correspondence as a plausible hypothesis for future work rather than a proven causal mechanism.

6.3 CROSS-PROFILE INTERPRETATION OF MULTI-PERSON BEHAVIOR

The central interpretive result of the thesis emerges only when all three architectural profiles are considered together. Detection count alone does not explain coach-scene robustness.

The MediaPipe–YOLOv8 comparison is partially consistent with detection-level behavior. YOLOv8 exposes two or more persons in 62% of sampled coach frames, compared with 23% for MediaPipe. This creates more opportunities for downstream misassignment, and Figure 4

is consistent with that interpretation: the same frame yields a single MediaPipe pose and multiple YOLOv8 detections.

The MediaPipe–MoveNet comparison is different. MoveNet exposes multiple persons in only 16% of sampled coach frames—even fewer than MediaPipe’s 23%—yet its coach-scene outlier rate is still higher. Detection counts therefore do not account for this difference. The residual mechanism must lie beyond simple secondary-person frequency, potentially in candidate retention, thresholding, or post-detection selection behavior. Panel (c) of Figure 4 illustrates one such instance, where MoveNet detects both persons but its largest-area selection places the native bounding box on the therapist rather than on the patient. This is precisely a selection-stage mechanism: detection is not the bottleneck in that frame. The current framework does not expose those internal states, so this mechanism cannot be localized further.

The MoveNet–YOLOv8 comparison makes the same point from another angle. The two models differ by roughly a factor of four in secondary-person detection frequency (16% vs. 62%), yet their coach-scene outlier rates are nearly identical (13.8% vs. 13.9%). The combined evidence from all three pairings therefore rules out a simple monotonic relationship between “how often additional persons are exposed” and “how often coach-scene failures occur”.

This is the strongest mechanistic conclusion that the present data support. Part of the MediaPipe–YOLOv8 difference is consistent with detection-level behavior; the MediaPipe–MoveNet difference remains only partially explained; and the full three-model comparison shows that detection count alone is insufficient as a global explanation of coach-scene robustness.

6.4 HIP-SPECIFIC INTERPRETATION

The hip sensitivity analysis provides a clear model-specific result. MediaPipe’s hip-region error is substantially higher than that of MoveNet and YOLOv8, and removing the hip joints reduces the MediaPipe–MoveNet gap from 2.01 percentage points to 1.21 percentage points. The remaining gap shows that the hip joints are not the only source of difference, but they account for a large share of it.

This pattern is consistent with a landmark-definition mismatch rather than a uniform degradation of MediaPipe’s full-body predictions. MediaPipe does not underperform uniformly across all regions; the discrepancy is concentrated in a structurally specific part of the skeleton.

That makes a body-model or landmark-definition explanation more plausible than a generic “lower quality” explanation, although the current evidence remains indirect.

6.5 CROSS-DATASET INTERPRETATION

Across the nine-model set, the ranking shift is real, and it persists even when COCO is restricted to single-person images. Crowded scenes alone do not appear sufficient to explain the shift.

At the same time, the comparison is protocol-bound. COCO differs from REHAB24-6 in target matching and visibility handling, so the result does not justify a blanket claim that standard benchmarks are invalid for rehabilitation-oriented deployment. What it does justify is a narrower conclusion: benchmark rankings transfer only partially under the evaluation protocol used here. In particular, the strongest REHAB24-6 configuration is not the strongest COCO configuration, and vice versa.

Among the three main models, the cross-dataset shift is also more nuanced than the full nine-model view. MoveNet remains the strongest of the three on both datasets, while MediaPipe and YOLOv8 swap order between REHAB24-6 and COCO. This supports limited transferability rather than wholesale benchmark failure.

6.6 COMPARISON WITH RELATED WORK

REHAB24-6 BENCHMARK STUDIES. Černek et al. [ČSB25] introduced REHAB24-6 and evaluated selected pose-estimation methods on the same benchmark, but did not include MoveNet and did not analyze temporal stability, coach-scene failure behavior, or the interaction between person exposure and wrong-person failures. The present thesis extends that benchmark work in all three respects.

CLINICAL MOVEMENT ANALYSIS. Rode et al. [Rod+25] evaluated a larger set of pose-estimation models for clinical movement analysis. Our work is narrower in model scope but deeper in deployment-oriented characterization: it adds mobile-specific speed–accuracy trade-offs, frame-to-frame prediction-displacement analysis, and explicit coach-scene failure analysis. The two perspectives are therefore complementary.

VIEWPOINT EFFECTS. Prior work already established that viewpoint affects HPE performance [Agu+23; BRS25]. The present thesis confirms that result for mobile models and further shows that viewpoint sensitivity, detection completeness, and coach-scene failure behavior do not collapse into a single robustness ranking.

MOBILE MODEL COMPARISONS. Previous mobile HPE comparisons focused mainly on speed and general benchmark accuracy [JK22; COL22]. The contribution here is not simply that three popular models were compared again, but that they were compared under a more decision-relevant set of dimensions for rehabilitation-oriented deployment.

6.7 PRACTICAL RECOMMENDATIONS

The recommendations follow directly from the bounded claims above.

DEFAULT CHOICE FOR BODY-JOINT ANALYSIS. If the application uses only the shared body-joint skeleton and runs under CPU constraints, MoveNet MultiPose is the strongest default among the three main models. It achieved the lowest REHAB24-6 error, the lowest main-model prediction displacement, and much higher throughput than MediaPipe Full.

WHEN SPEED DOMINATES. If throughput is the dominant constraint and some loss in body-joint accuracy is acceptable, MoveNet_SP_Lightning is the fastest evaluated configuration by a wide margin. It should be chosen only when that speed gain matters more than its weaker REHAB24-6 accuracy.

WHEN DETECTION COMPLETENESS DOMINATES. If downstream processing depends strongly on obtaining all 12 evaluated joints, YOLOv8 remains attractive because it produces complete detections most frequently. That benefit should be weighed against its weaker viewpoint robustness and its higher coach-scene failure rate relative to MediaPipe.

WHEN RICHER LANDMARK OUTPUT MATTERS. MediaPipe remains attractive for applications that benefit from richer native landmarks or simpler API integration. Its body-joint benchmark performance is weaker than MoveNet’s, but its coach extreme-case failure rate is lower and its native anatomical output is broader.

DEPLOYMENT GUIDANCE. Applications should provide frontal camera-position guidance, monitor for secondary-person interference, and surface quality warnings when predictions become incomplete or unstable. The results suggest that these deployment decisions can matter as much as the model choice itself.

6.8 LIMITATIONS

Several limitations bound the interpretation of these results.

BENCHMARK POPULATION. The central limitation is the benchmark itself. REHAB24-6 contains healthy adult volunteers rather than a real patient cohort. Although the recordings are physiotherapist-guided and exercise-specific, they do not capture the full movement variability, compensatory behavior, or impairment-specific patterns of clinical home-rehabilitation populations. The dataset authors explicitly note that the inclusion of real patient data would provide substantially greater clinical relevance. The current conclusions should therefore be read as evidence from a rehabilitation-oriented benchmark, not as direct clinical validation.

PRIMARY INFERENTIAL UNIT. The repository-state subject aggregation relies on the segmentation metadata `person_id`, yielding 10 primary subject clusters. This is a defensible in-repo clustering variable, but it remains a small inferential basis. The cluster-level statistics are much stronger than treating frames as independent observations, yet they are still bounded by a limited number of subjects.

COACH EXTREME-CASE SUBSET. Only five c17 views form the coach subset, and the mechanistic interpretation relies on external behavior rather than internal detector traces. This is sufficient to distinguish the MediaPipe-YOLOv8 and MediaPipe-MoveNet cases, but not to fully localize the residual mechanism behind the latter.

FRAME-INDEPENDENT EVALUATION. All models were evaluated frame-independently by design. This isolates model-level behavior, but it also means that the reported person-switch and temporal-stability values describe unsmoothed outputs and should be interpreted as upper-bound failure characteristics rather than as the performance of a fully engineered deployment pipeline.

TEMPORAL-STABILITY PROXY. The frame-to-frame prediction-displacement metric does not subtract ground-truth motion. It is useful for comparing how much the predicted trajectories move under the shared protocol, but it cannot isolate model noise from real subject motion or from model smoothing.

CROSS-DATASET PROTOCOL DIFFERENCES. The COCO comparison uses a different protocol from REHAB24-6, including target matching and visibility filtering. The ranking shift is therefore informative about limited transferability under the present protocol, but it is not a clean causal estimate of domain shift.

CPU-ONLY BENCHMARK. The inference benchmark was conducted on CPU-only desktop hardware. Relative rankings are useful for deployment trade-offs, but mobile-device acceleration may change absolute latency and possibly the ordering of some variants.

2D EVALUATION ONLY. The thesis evaluates 2D joint localization. Many rehabilitation applications ultimately depend on 3D joint angles, depth, or movement quality in 3D. The present results therefore characterize an important prerequisite, but not the full downstream assessment problem.

6.9 FUTURE WORK

Real patient validation is the most important next step. Four further directions follow.

REAL PATIENT COHORTS. The most important extension is validation on actual rehabilitation patients. That is the clearest path to greater clinical relevance and to testing whether the present ranking persists under impairment-specific movement patterns.

MECHANISTIC MULTI-PERSON ANALYSIS. Future work should log or expose internal candidate detections, detector thresholds, and retained instances for multi-person scenes. That would allow the MediaPipe–MoveNet asymmetry to be studied directly rather than inferred from external behavior.

STANDARDIZED CROSS-DATASET PROTOCOL. A cleaner cross-dataset evaluation would align target matching and visibility handling across benchmarks. This would strengthen claims about ranking transferability and reduce protocol-induced ambiguity.

MOBILE-DEVICE DEPLOYMENT. The speed benchmark should be repeated on representative smartphones or tablets, ideally with hardware-specific delegates enabled. That would convert the current relative CPU comparison into a directly deployment-facing benchmark.

UNIFORM TEMPORAL POST-PROCESSING. All models were evaluated frame-independently. A subsequent study could apply the same temporal post-processing or tracking layer to all models and quantify how much of the remaining instability is reducible at pipeline level.

7 CONCLUSION

This thesis evaluated three mobile pose estimation models and six additional variants on a rehabilitation-oriented proxy benchmark to study how well standard benchmark rankings carry into rehabilitation-oriented use. The evaluation covered body-joint accuracy, temporal stability, viewpoint sensitivity, multi-person behavior, detection completeness, inference speed, and cross-dataset transferability.

Two findings stand out. First, on REHAB24-6, MoveNet MultiPose is the strongest main-model default for body-joint analysis: it achieves the lowest subject-cluster error and the lowest frame-to-frame prediction-displacement value. Second, the three models do not collapse into a single ranking. YOLOv8 is the most detection-complete but also the most exposed to downstream disambiguation, MoveNet is the most conservative in secondary-person exposure yet does not show the lowest coach-scene failure rate, and MediaPipe shows the lowest catastrophic-failure rate on the coach extreme-case subset despite lower body-joint accuracy. The data therefore support three architectural profiles rather than a universal winner, and secondary-person detection frequency alone does not explain coach-scene robustness. A failure-mode decomposition reinforces this: each main model is dominated by a distinct failure category consistent with its architectural paradigm (Keypoint-Displacement for MediaPipe, Confidence-Collapse for MoveNet, Missing-Detection for YOLOv8).

The results should still be read within three limits. REHAB24-6 contains ten healthy adult volunteers performing physiotherapist-guided rehabilitation exercises rather than a clinical patient cohort. The cross-dataset comparison on COCO val2017 uses a different evaluation protocol, so the ranking shift should be read as limited transferability rather than as evidence of benchmark invalidity. All models were evaluated frame-independently; the reported person-switch and temporal-stability values therefore describe unsmoothed model outputs rather than a fully engineered deployment pipeline.

For CPU-based body-joint rehabilitation analysis, MoveNet MultiPose is the strongest default among the three main models. Other configurations are preferable when different criteria dominate: YOLOv8 for detection completeness, MediaPipe for lower coach-scene failure rate,

7. CONCLUSION

richer 33-landmark output, or simpler integration, and MoveNet_SP_Lightning for maximum throughput. Moving from this proxy benchmark toward clinical rehabilitation use will require validation on real patient data and cleaner cross-dataset protocols.

A SUPPLEMENTARY DATA

A.1 PER-EXERCISE ACCURACY

Table 16 reports mean NMPJPE per exercise and per main model on the REHAB24-6 valid subset. MoveNet ranks first on all six exercises; the MediaPipe–YOLOv8 ordering varies by exercise.

TABLE 16.: Mean NMPJPE (%) on the REHAB24-6 valid subset, per exercise and per main model. Bold marks the lowest error per row.

Exercise	MediaPipe	MoveNet	YOLOv8
Ex1	12.12	9.95	10.57
Ex2	11.66	9.00	11.07
Ex3	14.91	13.80	16.73
Ex4	12.32	11.39	13.62
Ex5	14.13	11.43	16.24
Ex6	11.89	10.10	12.23

A.2 FAILURE-MODE COUNTS

Table 17 reports the exact counts per failure category that underlie the distributions in Section 5.11. Percentages in parentheses are shares of each model’s total failures.

TABLE 17.: Failure-mode counts on REHAB24-6 for the three main models. Bold marks the dominant category per model.

Model	Failures	Miss.-Det.	Conf.-Coll.	Multi-Pers.	Kp.-Displ.
MediaPipe	1,583 (1.29%)	177 (11.2%)	135 (8.5%)	169 (10.7%)	1,102 (69.6%)
MoveNet	1,875 (1.53%)	369 (19.7%)	964 (51.4%)	38 (2.0%)	504 (26.9%)
YOLOv8	5,610 (4.58%)	3,580 (63.8%)	779 (13.9%)	42 (0.7%)	1,209 (21.6%)

REFERENCES

- [Agu+23] AGUILAR-ORTEGA, Rafael; BERRAL-SOLER, Rafael; JIMÉNEZ-VELASCO, Isabel; ROMERO-RAMÍREZ, FRANCISCO J.; GARCÍA-MARÍN, Manuel; ZAFRA-PALMA, Jorge; MUÑOZ-SALINAS, Rafael; MEDINA-CARNICER, Rafael & MARÍN-JIMÉNEZ, Manuel J.: “UCO Physical Rehabilitation: New Dataset and Study of Human Pose Estimation Methods on Physical Rehabilitation Exercises”. In: *Sensors* 23.21, 2023, p. 8862.
- [And+14] ANDRILUKA, Mykhaylo; PISHCHULIN, Leonid; GEHLER, Peter & SCHIELE, Bernt: “2D Human Pose Estimation: New Benchmark and State of the Art Analysis”. In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2014.
- [Baz+20] BAZAREVSKY, Valentin; GRISHCHENKO, Ivan; RAVEENDRAN, Karthik; ZHU, Tyler; ZHANG, Fan & GRUNDMANN, Matthias: “BlazePose: On-device Real-time Body Pose Tracking”. In: *CVPR Workshop on Computer Vision for Augmented and Virtual Reality*. 2020.
- [BRS25] BALDINGER, Melanie; REIMER, Lara Marie & SENNER, Veit: “Influence of the Camera Viewing Angle on OpenPose Validity in Motion Analysis”. In: *Sensors* 25.3, 2025, p. 799.
- [Cao+21] CAO, Zhe; HIDALGO, Gines; SIMON, Tomas; WEI, Shih-En & SHEIKH, Yaser: “OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43.1, 2021, pp. 172–186.
- [COL22] CHUNG, Jen-Li; ONG, Lee-Yeng & LEOW, Meng-Chew: “Comparative Analysis of Skeleton-Based Human Pose Estimation”. In: *Future Internet* 14.12, 2022, p. 380.
- [ČSB25] ČERNEK, Adam; SEDMIDUBSKÝ, Jan & BUDÍKOVÁ, Petra: “REHAB24-6: Physical Therapy Dataset for Analyzing Pose Estimation Methods”. In: *Similarity Search and Applications (SISAP 2024)*. Vol. 15268. Lecture Notes in Computer Science. Springer, 2025, pp. 18–33.

- [Deb+22] DEBNATH, Bappaditya; O'BRIEN, Mary; YAMAGUCHI, Motonori & BEHERA, Ardendu: "A review of computer vision-based approaches for physical rehabilitation and assessment". In: *Multimedia Systems* 28, 2022, pp. 209–239.
- [Gao+25] GAO, Zheyang; CHEN, Jiwei; LIU, Yonglong; JIN, Yuliang & TIAN, Duo: "A systematic survey on human pose estimation: upstream and downstream tasks, approaches, lightweight models, and prospects". In: *Artificial Intelligence Review* 58.3, 2025, p. 68.
- [Goo21] GOOGLE: *MoveNet: Ultra fast and accurate pose detection model*. <https://blog.tensorflow.org/2021/05/next-generation-pose-detection-with-movenet-and-tensorflowjs.html>. 2021.
- [Hel+21] HELLSTEN, Thomas; KARLSSON, Jonny; SHAMSUZZAMAN, Muhammed & PULKKIS, Goran: "The Potential of Computer Vision-Based Marker-Less Human Motion Analysis for Rehabilitation". In: *Rehabilitation Process and Outcome* 10, 2021.
- [Hii+23] HII, Chang Soon Tony; GAN, Kok Beng; ZAINAL, Nasharuddin; MOHAMED IBRAHIM, Norlinah; AZMIN, Shahrul; MAT DESA, Siti Hajar; WARRENBURG, Bart van de & YOU, Huay Woon: "Automated Gait Analysis Based on a Marker-Free Pose Estimation Model". In: *Sensors* 23.14, 2023, p. 6489.
- [Ion+14] IONESCU, Catalin; PAPAUA, Dragos; OLARU, Vlad & SMINCHISESCU, Cristian: "Human3.6M: Large Scale Datasets and Predictive Methods for 3D Human Sensing in Natural Environments". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 36.7, 2014, pp. 1325–1339.
- [Jac+10] JACK, Kirsten; MCLEAN, Sionnadh Mairi; MOFFETT, Jennifer Klaber & GARDINER, Eric: "Barriers to treatment adherence in physiotherapy outpatient clinics: A systematic review". In: *Manual Therapy* 15.3, 2010, pp. 220–228.
- [JCQ23] JOCHER, Glenn; CHAURASIA, Ayush & QIU, Jing: *Ultralytics YOLOv8*. 2023. URL: <https://github.com/ultralytics/ultralytics>.
- [JK22] Jo, BeomJun & KIM, SeongKi: "Comparative Analysis of OpenPose, PoseNet, and MoveNet Models for Pose Estimation in Mobile Devices". In: *Traitement du Signal* 39.1, 2022, pp. 119–124.
- [Lin+14] LIN, Tsung-Yi; MAIRE, Michael; BELONGIE, Serge; HAYS, James; PERONA, Pietro; RAMANAN, Deva; DOLLÁR, Piotr & ZITNICK, C. Lawrence: "Microsoft COCO: Common Objects in Context". In: *European Conference on Computer Vision (ECCV)*. Springer, 2014, pp. 740–755.

- [Lug+19] LUGARESI, Camillo et al.: *MediaPipe: A Framework for Building Perception Pipelines*. 2019. arXiv: [1906.08172](https://arxiv.org/abs/1906.08172).
- [NYD16] NEWELL, Alejandro; YANG, Kaiyu & DENG, Jia: “Stacked Hourglass Networks for Human Pose Estimation”. In: *Computer Vision – ECCV 2016*. Springer, 2016, pp. 483–499.
- [Rod+25] RODE, David; DUNKEL, Annika; WILLI, Romina; WOLF, Peter; XILOYANNIS, Michele & RIENER, Robert: “Assessment of monocular human pose estimation models for clinical movement analysis”. In: *Scientific Reports* 15, 2025, p. 38767.
- [Rog+24] ROGGIO, Federico; TROVATO, Bruno; SORTINO, Martina & MUSUMECI, Giuseppe: “A comprehensive analysis of the machine learning pose estimation models used in human movement and posture analyses: A narrative review”. In: *Heliyon* 10.21, 2024, e39977.
- [SRR24] SIMMICH, Joshua; ROSS, Megan H. & RUSSELL, Trevor: “Real-time video telerehabilitation shows comparable satisfaction and similar or better attendance and adherence compared with in-person physiotherapy: a systematic review”. In: *Journal of Physiotherapy* 70.3, 2024, pp. 181–192.
- [Sun+19] SUN, Ke; XIAO, Bin; LIU, Dong & WANG, Jingdong: “Deep High-Resolution Representation Learning for Human Pose Estimation”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, pp. 5693–5703.
- [TS14] TOSHEV, Alexander & SZEGEDY, Christian: “DeepPose: Human Pose Estimation via Deep Neural Networks”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2014, pp. 1653–1660.
- [Ull+25] ULLAH, Rahmat; ASGHAR, Ikram; NAWAZ, Rab; STACEY, Craig; AKBAR, Saeed & BISHOP, Peter: “A real time action scoring system for movement analysis and feedback in physical therapy using human pose estimation”. In: *Scientific Reports* 15, 2025, p. 44784.
- [XWW18] XIAO, Bin; WU, Haiping & WEI, Yichen: “Simple Baselines for Human Pose Estimation and Tracking”. In: *Computer Vision – ECCV 2018*. Springer, 2018, pp. 472–487.
- [Zen+22] ZENG, Ailing; YANG, Lei; JU, Xuan; LI, Jiefeng; WANG, Jianyi & XU, Qiang: “SmoothNet: A Plug-and-Play Network for Refining Human Poses in Videos”. In: *Computer Vision – ECCV 2022*. Springer, 2022, pp. 625–642.

- [Zhe+23] ZHENG, Ce; WU, Wenhan; CHEN, Chen; YANG, Taojiannan; ZHU, Sijie; SHEN, Ju; KEHTARNAVAZ, Nasser & SHAH, Mubarak: “Deep Learning-Based Human Pose Estimation: A Survey”. In: *ACM Computing Surveys* 56.1, 2023, pp. 1–37.

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